



SbC1, an R2R3-MYB transcription factor, specifically regulates anthocyanin accumulation in sorghum coleoptiles

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Abstract

Key message This study first identifies the SbC1, an R2R3-MYB transcription factor, specifically function as the key positive regulator for anthocyanin biosynthesis in sorghum coleoptiles.

Abstract Anthocyanins are pivotal in plant growth, development, and responses to biotic and abiotic stresses. However, the molecular mechanisms underlying anthocyanin biosynthesis in sorghum, one of the major cereal crops worldwide, remain largely unexplored. Here, through genome-wide association study (GWAS), virus induced gene silencing (VIGS) experiment and haplotype analysis, we identified a key R2R3-MYB gene, *SbC1*, that specifically regulates anthocyanin accumulation in sorghum coleoptiles but not in grain. Further transcriptomic analysis of the coleoptiles of the cultivars HYZ (*SbC1*) and QKY (*sbcl-a* mutant allele) demonstrated the positive regulatory role of *SbC1* in anthocyanin biosynthesis genes. The SbC1 protein predominantly localizes within the cell nucleus, where it interacts with Tan1. The interaction between SbC1 and Tan1 was confirmed through split-luciferase (Split-LUC), yeast two-hybrid (Y2H), and coimmunoprecipitation (Co-IP) assays. Comparative genomic analysis suggested that the R2R3-MYB transcription factors responsible for anthocyanin biosynthesis exhibit a similar molecular genetic basis in the parallel evolution of organ decoloration across different cereals. Additionally, overexpression of *SbC1* in the rice *Osc1* mutant complete rescue the anthocyanin accumulation defection and enhanced drought resistance compared with the control. In summary, for the first time, we identified the key transcription factor that specifically governs anthocyanin biosynthesis in sorghum coleoptiles. This discovery represents a significant breakthrough in understanding the molecular mechanisms of anthocyanin accumulation in sorghum and offers valuable genetic resources for plant breeding and biotechnology.

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Introduction

Anthocyanins are water-soluble pigments that impart red, blue, and purple hues to plants and possess significant antioxidant properties (Grotewold 2006; Das et al. 2012; Saigo et al. 2020). These properties increase plant tolerance

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to various biotic and abiotic stresses, such as intense light, drought, and low temperatures, thereby improving adaptability (Das et al. 2012; An et al. 2020a, 2020b). Moreover, anthocyanin accumulation in plant organs (e.g., stigmas, apiculi, coleoptiles, stems, roots, leaves, glumes and grains) serves as an efficient visual genetic marker, facilitating rapid assessment in functional gene research and crop breeding. For example, purple endosperm and coleoptiles are widely used as stable genetic markers in maize and wheat haploid breeding (Chen et al. 2022; Kumar et al. 2022).

The anthocyanin biosynthetic pathway has been extensively characterized, and structural genes involved in this pathway have been successfully isolated from a wide range of plant species (Dixon et al. 2013; Zhang et al. 2014). The genes in the anthocyanin synthesis pathway are regulated primarily by MBW complexes, which consist of R2R3-MYB transcription factors from subgroups 5 and 6, basic helix-loop-helix (bHLH) transcription factors from subgroup III_f, and WD-repeat proteins, or are regulated solely by MYB transcription factors from subgroup 7 (Zhang et al. 2014; Xu et al. 2015; Cappellini et al. 2021). The MYB transcription factor in the MBW ternary complex directly binds to the promoters of structural genes such as chalcone synthase (CHS), dihydroflavonol 4-reductase (DFR), anthocyanidin synthase (ANS), and glutathione S-transferase (GST), regulating the spatiotemporal expression of these genes and inducing anthocyanin accumulation in various tissues and organs (Xu et al. 2015; Wang et al. 2020; He et al. 2023a).

A transcription factor featuring a conserved R2R3-MYB domain regulates anthocyanin biosynthesis in crops such as maize and rice. In maize, the gene encoding the transcription factor is represented by *ZmC1* (*colorless 1*) and its duplicate *ZmPL1* (*purple leaf 1*), and its homologue in rice is *OsC1*. The *ZmC1* gene in maize was the first identified R2R3-MYB gene that regulates anthocyanin accumulation in plants. It specifically controls pigment accumulation in the aleurone and scutellum of maize kernels without affecting pigment accumulation in other plant parts (Paz-Ares et al. 1987; Chatham et al. 2019). Another transcription factor, *ZmPL1*, determines pigment accumulation in the vegetative and reproductive organs of maize, such as leaves, glumes, and anthers (Zhang et al. 2000; Petroni et al. 2014; Chatham et al. 2019). The regulatory genes involved in anthocyanin biosynthesis in rice were identified by comparison with the sequences of known maize orthologues, genome-wide association studies (GWASs), and transcriptomic analysis (Zheng et al. 2019, 2021). At least nine anthocyanin regulators that positively control the colouration of plant organs, including two R2R3-MYBs (*OsC1* and *OsMYB3/Oskala3*), have been identified in rice (Sakamoto et al. 2001; Fan et al. 2008; Zheng et al. 2019, 2021; Kim et al. 2021; Meng et al. 2021; Xia et al. 2021; Yang et al. 2021). *OsC1* is responsible for pigmentation in vegetative and reproductive organs,

including the leaf sheath, apiculus, hull, and stigma (Fan et al. 2008; Sun et al. 2018; Du et al. 2022; Jiang et al. 2024). *OsMYB3/Oskala3* is a key regulator that determines pigment accumulation in rice pericarps (Kim et al. 2021; Zheng et al. 2021). In other cereal crops, such as foxtail millet and wheat, the *C1* homologue genes *SiC1* and *TaC1* regulate anthocyanin accumulation in vegetative organs/tissues, respectively (Jiang et al. 2018; Li et al. 2022).

Sorghum, the fifth major cereal crop globally, is known for its diverse plant, glume, and grain colours, which are attributed to various flavonoids, such as anthocyanins and tannins (proanthocyanins). These compounds not only promote human health and reduce damage caused by birds but also play crucial roles in the formation of aromatic substances in sorghum distilled liquor (Dykes 2019; Zhang et al. 2023; Ding et al. 2024). Research on the genetic mechanisms underlying anthocyanin and tannin biosynthesis in sorghum has been a significant focus globally (Wu et al. 2012, 2019; Dykes 2019; Xie et al. 2019; Zhang et al. 2023; Ding et al. 2024). Numerous studies have reported quantitative trait loci (QTLs) for traits related to anthocyanin and tannin synthesis in sorghum, such as grain colour, tannin content, and plant colour, using both biparental QTL mapping and GWAS to identify significant trait-associated loci (Wu et al. 2012, 2019; Rhodes et al. 2014; Girma et al. 2019; Mace et al. 2019; Xie et al. 2019; Yang et al. 2024). Boddu et al. performed fine mapping of QTLs controlling sorghum grain colour and identified the MYB transcription factor Y1 as a candidate gene that regulates pigment deposition and 3-deoxyanthocyanidin accumulation, affecting grain colour and mould resistance (Boddu et al. 2005; Nida et al. 2021; Tao et al. 2021a, b). *Tannin1* (*Tan1*) and *Tannin2* (*Tan2*) encode a WD40 protein and a bHLH protein, respectively, and have been confirmed as pivotal genes involved in anthocyanin and tannin biosynthesis (Wu et al. 2012, 2019). Recent studies have shown that introducing the sorghum MYB transcription factor SbTT2 into maize activates the anthocyanin and tannin biosynthesis pathways (Lu et al. 2023). Despite extensive researches have been done, the number of identified genes for sorghum anthocyanin and tannin biosynthesis remains limited, and studies on the molecular mechanisms of key candidate genes are scarce.

The coleoptile is a key protective tissue in monocot seedlings (Inada et al. 2002). In sorghum, anthocyanin accumulation within the coleoptile neutralizes reactive oxygen species, reducing oxidative damage. This enhances seedling survival under drought, UV radiation, salinity, and other stress conditions, improving stress tolerance and adaptability (Yang et al. 2024). The objective of this study is to identify key regulatory genes governing anthocyanin accumulation in sorghum through GWAS and to elucidate their molecular mechanisms. In this study, we discovered that the *SbC1* gene plays a crucial role in regulating anthocyanin accumulation

in sorghum coleoptiles by performing GWAS, collinearity analysis and virus induced gene silencing (VIGS) experiment. Haplotype analysis revealed that the allelic variant *sbcl-a* of the *SbC1* gene is associated with anthocyanin synthesis in the coleoptile but does not affect anthocyanin or tannin synthesis in the grain. Additionally, we found that *SbC1* interacts with *Tan1* and that *SbC1* regulates the expression of structural genes, such as *F3H*, *DFR*, and *ANS*, involved in anthocyanin biosynthesis, thereby controlling anthocyanin accumulation in sorghum coleoptiles. Overexpression of *SbC1* in a rice *OsC1* mutant restored anthocyanin accumulation in tissues such as leaf sheaths, glumes, and stigmas, and significantly enhanced drought resistance. Collectively, our study has identified a pivotal gene positively regulates anthocyanin biosynthesis specifically in sorghum coleoptile, thereby providing a molecular basis and candidate gene for breeding stress-resistant varieties.

Materials and methods

Plant materials and phenotypic assessment

A collection of 242 sorghum germplasm lines, primarily comprising of landrace cultivars, modern cultivars, and advanced breeding materials sourced from China, were utilized in this study (Table S1). These resources were procured from the Center for Crop Germplasm Resources, Institute of Crop Sciences, Chinese Academy of Agricultural Sciences (Beijing, China), and the Institute of Upland Crops, Guizhou Academy of Agricultural Sciences (Guiyang, China). In 2022, a phenotypic evaluation trial was conducted in Guiyang, Guizhou, China, to examine the coleoptile colour of the population. Germination assays were performed in Petri dishes, with seeds treated with a 1% hydrogen peroxide solution for 16 h, followed by rinsing with sterile water and culture on water-saturated filter paper. Coleoptile colour was assessed and recorded during the 7- to 10-day postgermination period: green was denoted as 1, and red was denoted as 2.

Anthocyanin content determination

The method of Wang et al. (2022) was used to measure total anthocyanin content. A 0.1 g leaf sheath from 2-week-old sorghum seedlings was placed in a 1.5 ml centrifuge tube with 800 µl of 0.1% HCl in methanol. The sample was ground and incubated at 4 °C for 1 h, with vortexing every 15 min (3–5 times). After centrifuging at 12,000 g for 8 min at 4 °C, the supernatant was collected. The residue was re-extracted three more times with the same solvent. The supernatants were combined and filtered through a 0.22 µm membrane into a new 10 ml tube. A 200 µl aliquot was

diluted tenfold, and absorbance was measured at 530 nm and 657 nm. Total anthocyanin content was calculated using the formula: Total anthocyanin content = $(A_{530} - 0.25 \times A_{657}) / M$, where M is the fresh weight of the sample (g). Each sample was analyzed in triplicate.

GWAS

Genomic resequencing and SNP discovery were performed on 242 sorghum germplasm lines on the basis of our previous research (Zhang et al. 2023). The reference genome sequence and annotation files of sorghum (BTx623, v3.1.1) (McCormick et al. 2017) were acquired from the Phytozome database (<https://phytozome-next.jgi.doe.gov/>) (Goodstein et al. 2012). Quality control was executed on the raw reads via fastp software (v0.20.0) (Chen et al. 2018), resulting in clean reads. Subsequent alignment with the reference genome was performed via the BWA-MEM algorithm within BWA software (v0.7–17) (Li and Durbin 2009). SNPs for each genotype were identified and filtered via the Select-variants and VariantFiltration tools within GATK4 software (v3.8.1) (McKenna et al. 2010), followed by further filtering with VCFtools software (v0.1.17) (Danecek et al. 2011) under the following criteria: retaining only biallelic variants, minimum allele frequency > 5%, and missing rate < 20%, ultimately resulting in 2,015,850 high-quality SNPs. The GWAS for coleoptile colour was executed employing the GLM and Blink statistical methods within GAPIT software (v3.0) (Wang et al. 2021). In the mixed linear model, the first three principal components derived from the whole-genome SNPs were designated as fixed effects to account for population structure, with random effects clustered on the basis of the genetic relationships among all genotypes. The genome-wide significance threshold was determined via the adjusted Bonferroni correction threshold, set at $P = -\log_{10}(0.05/2,015,850) = 7.6$.

Identification of candidate genes

LD analysis and inference of haplotype blocks surrounding peak SNPs were conducted with LD BlockShow software (v1.39) (Dong et al. 2021). On the basis of pairwise R^2 values among SNPs within the localized Manhattan block region on chromosome 10, genomic regions encompassing candidate genes were delineated through LD, aiming to identify genes potentially influencing the traits of interest. The genomic sequence of HYZ v1.0 was retrieved from the China National GeneBank at <https://db.cngb.org/search/project/CNP0002968/> (Ding et al. 2024). Collinearity relationships among the sorghum genomes BTx623 and HYZ within the candidate intervals were analysed via minimap2 software (v2.26) (Li 2021), and visual representations were

generated in conjunction with the genomic annotation results via NGenomeSyn software (v1.41) (He et al. 2023, c).

Phylogenetic analysis and comparative mapping

The protein sequence of the sorghum *SbC1* gene (*SbHYZ.10G81400*) was derived from the HYZ V1.0 reference genome. Protein sequences homologous to *SbC1* from other Gramineae crops were retrieved from the Phytozome database, followed by conserved domain analysis using the InterPro database. Multiple sequence alignments were performed via MAFFT software (v7.48) (Katoh et al. 2022). Phylogenetic trees were constructed via the maximum likelihood method, with MEGA software (v11.0) (Kumar et al. 2018), using 1000 bootstrap replicates. Genomic datasets for maize (B73, v4.0) (Wei et al. 2009), rice (Nipponbare, v7.0) (Ouyang et al. 2007), wheat (Chinese Spring, v2.1) (Zhu et al. 2021), millet (Yugu1, v2.2) (Bennetzen et al. 2012), and sorghum (HYZ, v1.0) were acquired from Phytozome. The collinearity relationships of gene sequences across diverse species genomes were analysed via Jcvi software (v1.2.4) (Tang et al. 2024), and visual representations of the collinearity maps were generated via R software (v4.21) in conjunction with genomic annotation. Genome comparisons were conducted using protein sequences based on the Diamond software (v2.1.7.161) (Buchfink et al. 2015).

Haplotype and population genetics analysis

Using the genome sequence of the *S. bicolor* variety HYZ as a reference and resequencing data from 479 global sorghum germplasm samples (Boatwright et al. 2022; Zhang et al. 2023) (Table S7), polymorphic SNP loci within the *SbC1* gene and its promoter sequence (2000 bp) were identified. Coupled with the geographical origin of the materials and the phenotypic traits of coleoptile colour, haplotype and haplotype network analyses of the candidate genes were conducted via CandiHAP software (v1.30) (Li et al. 2023). The phylogenetic tree of the *SbC1* gene was constructed with vcF2Dis software (v1.50) (<https://github.com/BGI-shenzhen/VCF2Dis>). Amplification and sequencing of the *SbC1* gene were executed through Sanger sequencing technology, with the assembled sequences analysed using DNASTAR Lasergene software (v11.0) (Burland 1999). Sequence alignment was performed using MAFFT software to ascertain the accuracy of critical variant loci.

Cloning, vector construction and transformation

The primers used were designed using Premier software (v5.0) (Lalitha 2000) (Table S8), and the full-length CDS of the *SbC1* gene was amplified from cDNA derived from the red coleoptile of the sorghum variety HYZ. After

amplification, the gene was cloned and inserted into the temporary vector PUC57 and, after sequencing verification, was ligated into the *pCAMBIA1300* vector with the maize ubiquitin promoter (UBI), resulting in the UBI construct (Figure S1). This construct was subsequently introduced into *Agrobacterium tumefaciens* strain EHA105, and positive colonies were selected after transformation into rice variety 9311, following the *Agrobacterium*-mediated mature seed transformation protocol (Toki et al. 2006). Transgenic rice lines were screened on Murashige and Skoog medium supplemented with 75 mg/L hygromycin. DNA was extracted from the rice leaf samples, and the overexpression lines were identified via PCR via specific primers (Table S8). The T₂ generation of positive rice lines was utilized for phenotypic analysis and anthocyanin quantification.

VIGS assay

The VIGS procedure was modified according to our previous report (He et al. 2023b). Sorghum seeds were sterilized (10% NaClO, 5 min; 75% ethanol, 3 min), rinsed (sterile ddH₂O, 5×), and germinated (filter paper, 2 days). A 300-bp *SbC1* fragment was amplified by PCR from HYZ seedling cDNA using specific primers (Table S8), digested (*Kpn* I, *Nco* I), and subsequently cloned and inserted into *pTRV2*, which yielded the *TRV2-SbC1* construct. The *pTRV1*, *pTRV2*, and *TRV2-SbC1* constructs were subsequently transformed into *A. tumefaciens* GV3101. For VIGS, *Agrobacterium* combinations (*pTRV1* + *pTRV2* or *pTRV1* + *TRV2-SbC1*) were prepared (1:1 ratio) and used for seed inoculation. Inoculation involved vacuum infiltration (100 µM AS, 3.3 mM Cys, 5 mg L⁻¹ Tween 20; ~95 kPa, 2 h). Inoculated plants were cultured (18 °C, 16 h light/8 h dark cycle).

Subcellular localization of SbC1

To determine the subcellular localization of *SbC1*, the CDSs of *SbC1* and *Tan1* were individually cloned and inserted into the *pCAMBIA1300-GFP* vector. The *agrobacteria* with the GFP fusion plasmids and *NLS-mCherry* (NLS, nuclear localization sequence) were coinfiltrated into the leaves of tobacco (*Nicotiana benthamiana*) plants. At 3 days after infection, fluorescence signals were observed via a Leica SP5 confocal microscope (Leica, Wetzlar, Germany).

BiFC assay

For the interaction between *SbC1* and *Tan1* in planta, the CDSs of *SbC1* and *Tan1* were cloned and inserted into the *pXY106* (nYFP) and *pXY104* (cYFP) vectors, respectively. The resulting plasmids, along with the empty vectors *pXY104* and *pXY106*, were introduced into *A. tumefaciens* strain GV3101, followed by transient expression in tobacco

leaves over a 3-day period with the specified combinations. The yellow fluorescent protein (YFP) fluorescence signal was detected via a Leica SP5 confocal microscope.

Split-LUC assay

The Split-LUC assay was performed as previously described (Zhou et al. 2018). First, the CDS of SbC1 and *sbc1-a* were cloned and inserted into the *pCAMBIA1300-nLUC* vector, whereas the CDS of Tan1 and *scb1-a* were cloned and inserted into the *pCAMBIA1300-cLUC* vector. The resulting plasmids were transiently coinfiltrated into tobacco leaves for 3 days with the indicated combinations. The leaves were briefly immersed in luciferin buffer for 2 min, and LUC signals were detected via a chemiluminescence apparatus (Tanon, 5200, Shanghai, China).

Y2H assays

To confirm the interaction between SbC1 and Tan1, the CDS of *Tan1* was cloned and inserted into the *pGADT7* vector, whereas the CDS of *SbC1* was cloned and inserted into the *pGBKT7* vector. Plasmids with various AD/BD combinations were transformed into the yeast strain AH109 according to the standard procedure. The resulting yeast transformants were then plated onto minimal SC/Trp-Leu medium or SC/Trp-Leu-His-Ade medium and grown for 3 days at 28 °C.

Co-IP assays and immunoblot analysis

Co-IP assays were performed as previously described (He et al. 2023a). To confirm the interaction between SbC1 and Tan1 in planta, the CDS of *SbC1* was cloned and inserted into the *pCAMBIA1300-GFP* vector, whereas the CDS of *Tan1* was cloned and inserted into the *pET101-Flag* vector. The resulting plasmids, as well as the *pCAMBIA1300-GFP* empty vectors, were transiently expressed in tobacco leaves for 3 days with the indicated combinations. Total proteins were extracted with IP buffer (10 mM Tris-HCl [pH 7.5], 150 mM NaCl, 2 mM EDTA, 1 mM MG132, 0.5% [v/v] Nonidet P-40) and incubated with anti-Flag antibody-conjugated agarose (Medical Biological Laboratories, M185-10, Nagoya, Japan). Proteins were then probed by immunoblot analysis using anti-Flag (Medical Biological Laboratories, M185-7) and anti-GFP (TransGen Biotech, HT801, Beijing, China) antibodies.

Transcriptomic analysis

Following haplotype analysis, two sorghum varieties bearing the *SbC1* allele, HYZ (*SbC1*) and QKY (*sbc1-a*), were delineated from 242 sorghum germplasms. These varieties

exhibit closely related phylogenetic relationships. RNA was extracted from the sheaths of three-leaf-stage plants via the TRIzol method. Upon quality validation, RNA samples were sequenced via the Illumina NovaSeq 6000 platform (Illumina, San Diego, California, USA). The raw data were processed via fastp software to filter out low-quality reads, thereby obtaining high-quality clean reads. These clean reads were aligned against the sorghum HYZ reference genome via HISAT2 software (Version 2.2.1) (Kim et al. 2019), enabling the acquisition of genomic location and sequence feature information specific to the sequenced samples. The fragments per kilobase of transcript per million mapped reads (FPKM) values were calculated, and gene and transcript expression levels were assessed. Differential gene expression analysis was performed via the EdgeR package (v3.20) (Robinson et al. 2010), with the selection criteria $|\log_2\text{-fold change}| > 1$ and $P \text{ value} < 0.05$. DEGs were annotated, and pathway enrichment analysis was conducted via the GO and KEGG databases.

qRT-PCR

Total RNA was extracted from the sheaths of sorghum varieties HYZ and QKY via the RNAPrep Pure Plant Kit (TIANGEN, Beijing, China). cDNA synthesis was performed via the PrimeScript™ RT Reagent Kit with gDNA Eraser (TaKaRa, Kusatsu, Japan) according to the manufacturer's instructions. qRT-PCR was carried out using PowerUp SYBR Green Mix (Thermo Fisher Scientific, Waltham, MA, USA). The reaction mixture comprised 5 µl of PowerUp SYBR Green reagent, 1.0 µl of cDNA template, 0.25 µl of each primer (10 µM), and 3.5 µl of nuclease-free water. The qRT-PCR was performed on an ABI7500 Real-Time PCR System (Thermo Fisher Scientific) following the manufacturer's guidelines, with the sorghum tubulin alpha gene (*Sobic.001G107200*) serving as the reference gene. Relative gene expression levels were calculated via the $2^{-\Delta\Delta C_t}$ method (Livak and Schmittgen 2001), and the sequences of the primers used for qRT-PCR are provided in Table S8.

Drought tolerance assays

Seeds from the transgenic rice lines, wild-type rice, HYQ, and QKY were used as experimental materials and were germinated in an artificial climate chamber under long-day conditions (28 °C, 16 h light/8 h dark). Two weeks after seedling emergence, seedlings of uniform height sorghum and rice seedlings were subjected to drought stress via 20% PEG-6000. Two weeks after the onset of stress, phenotypes were observed, and data on the fresh weight and survival rates of the plants before and after drought stress were collected.

Data processing and statistical analysis

Descriptive statistical analysis and Student's t-test were performed using SPSS software (v22.0), with a significance level set at $P < 0.05$.

Results

Identification of candidate genes regulating sorghum coleoptile colouration

The coleoptile plays a critical role in facilitating the emergence of sorghum grains from the soil. The accumulation of anthocyanins in the coleoptile provides protective effects on seedlings against diverse stresses. To identify quantitative trait loci (QTLs) for the colouration of sorghum coleoptiles, we investigated the coleoptile colour of 242 sorghum germplasm resources at the seedling stage, specifically between 7 and 10 days after germination. Of these, 133 varieties

presented a red coleoptile phenotype, whereas 109 presented a green coleoptile phenotype (Fig. 1a and Table S1). Consistent with the differences in colour, the average total anthocyanin content of red coleoptile sorghum accessions (0.65 mg/g) was significantly greater than that of green coleoptile sorghum accessions (0.05 mg/g) (Fig. 1a). In general, the accumulation degree of pigments of sorghum coleoptiles was positively associated with the total anthocyanin content. By performing a GWAS, we identified a single QTL region significantly associated with coleoptile colour, located on chromosome 10 between 6.41 and 6.45 Mb. (Fig. 1b and Table S2). Linkage disequilibrium (LD) heatmap further revealed that the SNP associated with coleoptile color showed strong linkage with adjacent SNPs within this QTL region (Fig. 1b). On the basis of the genomic annotation information for the reference genome BTx623 (V3.1.1), we identified three candidate genes (Sobic.010G077250, Sobic.010G077275, and Sobic.010G077300) within 50 kb vicinity of the significant single-nucleotide polymorphism (SNP) site. Protein sequence homology analysis of these

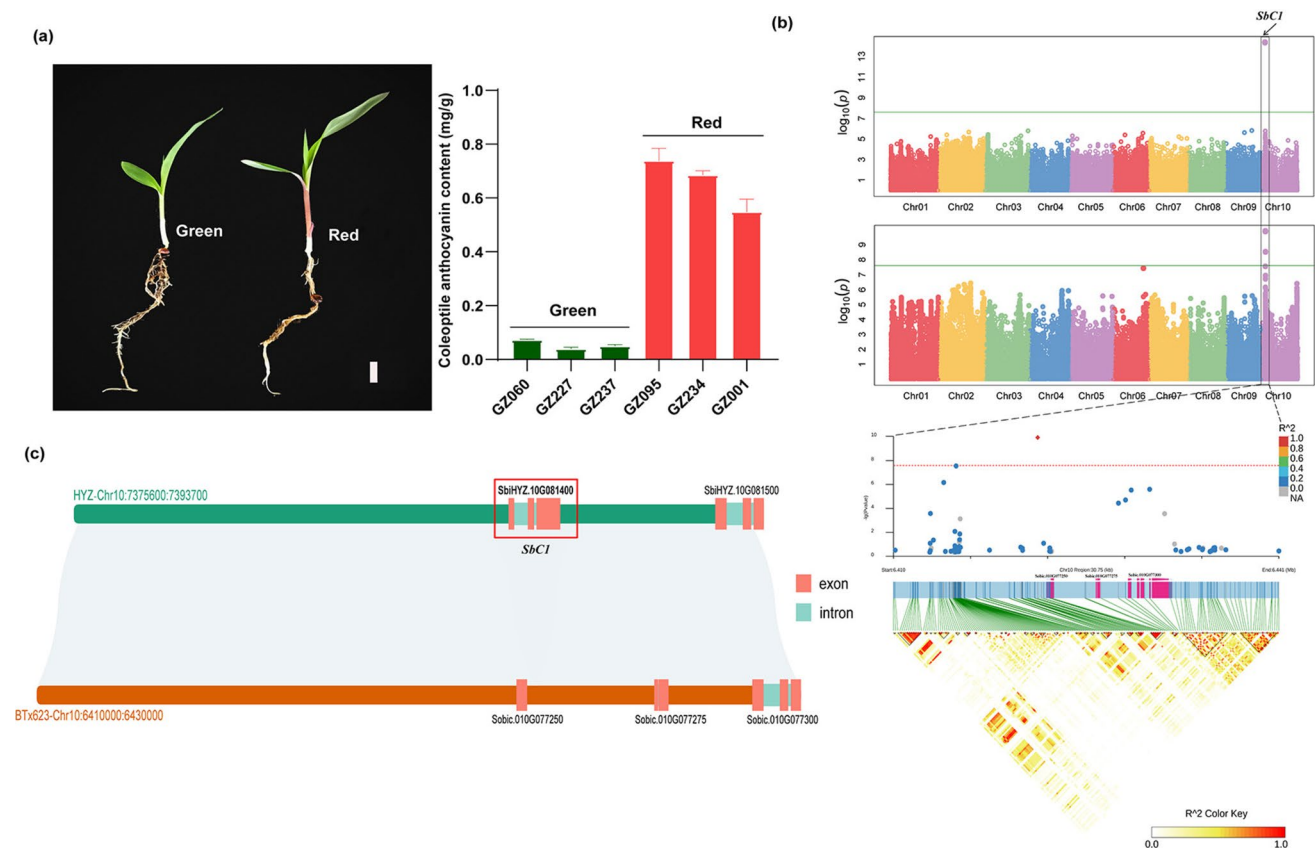


Fig. 1 Genome-wide association analysis of coleoptile colour in 242 sorghum accessions. **a** Coleoptile colour of green (left) and red (right) sorghum accessions; bars=1 cm. Total anthocyanin content in red- and green-coleoptile sorghum accessions. **b** Manhattan plots showing the results of the GWAS of green- and red-coleoptile sorghum accessions. The peak regions on chromosome 10 are displayed

in conjunction with the linkage disequilibrium blocks. The lines indicate the significance threshold value ($P < 2.48 \times 10^{-8}$). **c** Collinearity analysis was performed on the candidate region of chromosome 10 in the sorghum genomes HYZ (version v1.0) and BTx623 (version v3.1.1). The collinearity intervals are 7.37–7.39 Mb and 6.41–6.43 Mb

three genes revealed no similarity with previously cloned anthocyanin synthesis-related genes in *Arabidopsis*, maize, and rice (Table S3). Considering the possible lack of annotation on the anthocyanin biosynthesis pathway in the reference genome variety BTx623 (green coleoptile and white grain), we conducted a microcolinearity analysis by comparing the candidate region sequences of BTx623 with the genome sequences of the sorghum cultivar HongYingZi (HYZ, red coleoptile and brown grain) (Ding et al. 2024). This led to the identification of a candidate gene, SbiHYZ.10G081400, in the HYZ genome (Fig. 1c). Notably, this gene was absent in the BTx623 genome and is hereafter referred to as SbC1 (*Sorghum bicolor* purple coleoptile) (Fig. 1c). The annotation analysis revealed that the protein sequence of SbC1 encodes an R2R3-MYB transcription factor (Figure S2).

SbC1 positively regulates anthocyanin accumulation in sorghum coleoptiles

By using a gene expression database to analyse expression across various tissues and developmental stages of the sorghum cultivar HYZ (Ding et al. 2024), we observed that the expression of the *SbC1* gene peaked in the stem/coleoptile tissues during the five-leaf stage, which coincided with the stage when coleoptile colouration became prominent (Figure S3). To investigate whether *SbC1* is responsible for the accumulation of anthocyanins in sorghum coleoptiles, we examined the phenotypes of 479 sorghum accessions for which resequencing data was available. By utilizing the sorghum HYZ genome as a reference for haplotype analysis, we identified a total of 19 SNP sites within the *SbC1* gene, encompassing 14 in the promoter region, two in exons and two in introns, and one in the 3'-untranslated region (3'UTR) (Table S4). Among these, a critical SNP variation characterized by a 'C' to 'T' mutation at the 124th (Chr10:7,387,103) base of the first exon of *SbC1* was identified. SNP_{C124T} resulted in the conversion of the glutamine-encoding codon CAG with the stop codon TAG, leading to premature termination of translation and yielding a truncated protein that lacks normal functionality (Fig. 2a). After observing the phenotypes of 37 accessions harbouring this specific mutation with the SNP_{C124T} genotype, we found that all 37 accessions presented green pigmentation in their coleoptiles, which indicated that *SbC1* might be a critical regulator of anthocyanin accumulation in sorghum coleoptiles. Consequently, the genotype that affects anthocyanin accumulation in coleoptiles carrying SNP_{C124T} was designated the *sbc1-a* allele (Figure S4a). The identification of the *sbc1-a* allele was further supported by Sanger sequencing (Fig. 2b).

To determine the impact of *SbC1* mutation on anthocyanin accumulation in other sorghum tissues, we investigated the genotypic and phenotypic data of sorghum germplasm

resources (Zhang et al. 2023). The phenotypes of sorghum varieties harbouring *SbC1* and *sbc1-a* alleles were specifically observed. During seed maturation, the pericarp and seed coat of *sbc1-a* variant grains presented red colouration that was similar to that observed in HYZ (Figure S4b and c). Furthermore, analysis of anthocyanin contents confirmed the presence of anthocyanins in both the pericarp and seed coat of the variety QingKeYang (QKY, a genetically similar line of HYZ with the *sbc1-a* allele) (Figure S3d). Additionally, 29 out of 37 varieties carrying the *sbc1-a* allele presented the same pericarp and testa colour as the QKY variety did (Table S5). Taken together, the findings from these tissues and those from coleoptiles suggest that allelic variation in the *SbC1* gene specifically affects anthocyanin accumulation in sorghum coleoptiles.

To elucidate the functional role of the *SbC1* gene, we conducted a virus-induced gene silencing (VIGS) experiment using HYZ (with the *SbC1* genotype) as the experimental material. At approximately 2 days postinoculation, we observed notable lightening of the colouration of the coleoptiles in the experimental group (*pTRV: SbC1*) compared with that in the control group (*pTRV: 00*) (Fig. 2c). Additionally, we measured the transcript levels of SbC1 and the anthocyanin content in the coleoptiles in both the experimental and control groups. The results demonstrated successful knockdown of SbC1 expression in the coleoptiles within the experimental group, which corresponded with a significant reduction in anthocyanin content (experimental: 0.17 mg/g vs control: 0.69 mg/g) (Fig. 2d), implying that downregulation of *SbC1* expression contributes to the lightening of coleoptile colour in HYZ seedlings.

The *sbc1-a* allele is found exclusively in Chinese varieties that might have been introduced from Africa and other regions of Asia

To investigate the sequence diversity of *SbC1* and determine the distribution of the *sbc1-a* allele in the sorghum population, a haplotype network analysis was conducted, accounting for the geographic origin of 479 global sorghum resources. The analysis revealed 14 haplotypes (with variety numbers > 5) with distinct geographic distribution patterns (Fig. 3a). Most haplotypes exhibited a wide distribution across diverse geographic populations, except for Hap4, which occurred exclusively in Chinese sorghum resources carrying SNP_{C124T} (*sbc1-a*). Compared with the other populations, the African sorghum population presented higher levels of haplotype and nucleotide diversity. The haplotypes Hap2, Hap3, and Hap5 were found to be the most common in the North American sorghum population, whereas Hap1 and Hap4 were predominantly observed in Chinese germplasms, which exhibited close genetic relatedness with only the SNP_{C124T} mutant (Fig. 3a), suggesting the potential

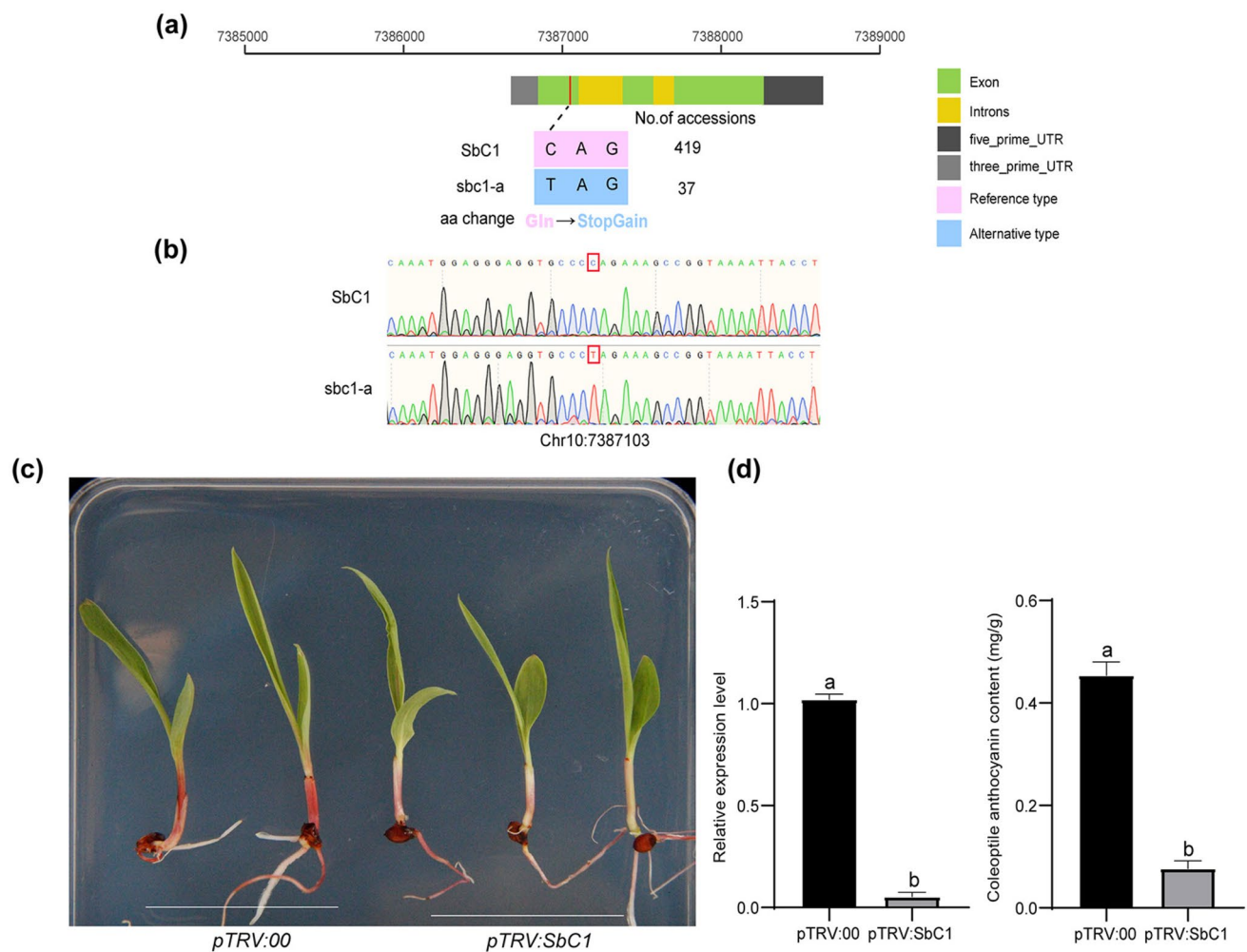


Fig. 2 Natural variation in *Sbc1* impacts anthocyanin accumulation in sorghum coleoptiles. **a** Important natural variation in *Sbc1*. Codon change from a glutamine (Gln) codon to a premature stop codon. **b** Alignment of the *Sbc1* nucleotide sequence to the *sbc1-a* sequence is shown. **c** Phenotypes of sorghum seedlings resulting from VIGS. **d**

Quantitative analysis of *Sbc1* gene expression and anthocyanin content in sorghum coleoptiles following VIGS. The error bars represent means $\pm$ SDs (n=3). Different lowercase letters indicate a significant difference at $P < 0.05$ based on a t-test

derivation of Hap4 from Hap1. Hap6, which is found mainly in Africa, North America, and India, was genetically classified into another subgroup that exhibited a closer genetic relationship to Hap1 (Fig. 3a). Thus, the haplotype Hap1 might have been introduced to China through the regions harbouring Hap1 and Hap6. Additionally, phylogenetic analysis of *Sbc1* gene diversity supported these findings, classifying different regional sorghum varieties into distinct branches, each fixing different haplotypes, with *sbc1-a*-carrying varieties clustered on a separate branch (Fig. 3b).

Comparative analysis of C1 locus in cereals

The annotation analysis revealed that the nucleotide sequence of *Sbc1* encodes an R2R3-MYB transcription factor. To identify the function of *Sbc1*, we first aligned its

amino acid sequence with those of other identified anthocyanin regulators in Poaceae crops. The results revealed that the protein sequence of *Sbc1* is gene codes for homologs to those of *ZmC1*, *ZmPL1*, *OsC1*, *SiC1*, and *TaC1-7A*, containing the canonical bHLH-binding motif [D/E]Lx₂[R/K]x₃Lx₆Lx₃R in its R2R3 domain in the N-terminal region and the characteristic motif sequence for subgroup 5 at its C-terminal region (Fig. 4a). The subsequent phylogenetic analysis revealed that the protein sequence of *Sbc1* clustered together with those of *ZmC1* and *ZmPL1*, indicating the homologous relationship of *Sbc1* with *ZmC1* and *ZmPL1* (Fig. 4b). A subcellular location analysis of the *Sbc1* protein revealed that, similar to *ZmC1* and *ZmPL1*, the *Sbc1* protein was located in the nucleus, suggesting its potential function as a transcription factor (Figure S5). Comparative mapping results indicate that the chromosomal region

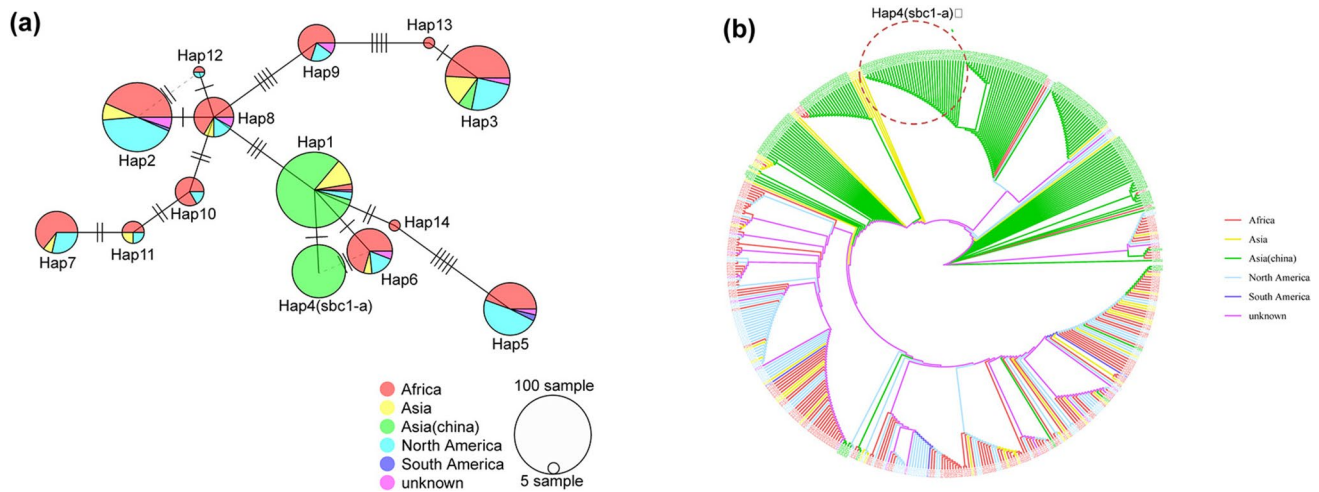


Fig. 3 Haplotype network analysis of *SbC1* from 479 sorghum accessions. **a** Haplotype network analysis of *SbC1*. Haplotypes of *SbC1* are represented by circles. The size of each circle is proportional to the frequency of the corresponding haplotype. Lines between haplo-

types represent the mutations, with each line representing one mutation; $N \geq 5$. **b** Phylogenetic tree of *SbC1* and its promoter region (2 kb)

containing the *SbC1* gene in sorghum corresponds to five highly conserved chromosomal segments, located on chromosome 4 in foxtail millet, chromosome 6 in rice, chromosomes 6 and 9 in maize, and chromosomes 7A, 7B, and 7D in wheat (Table S6). Genome micro-collinearity analysis also revealed significant differences in genome size and gene numbers among cereal species, yet their genomes exhibited evident collinearity at the chromosomal level, with conserved gene markers and gene order (Fig. 4c-f). These findings are consistent with previous studies (Lai et al. 2004; Li et al. 2022).

The *SbC1* protein physically interacts with Tan1

Previous studies have demonstrated that Tan1 (a homologue of *Arabidopsis* TTG1) and Tan2 (a homologue of *Arabidopsis* TT8) are responsible for condensed tannin accumulation in sorghum grains (Wu et al. 2012, 2019). In *Arabidopsis*, TTG1 and TT8 play critical roles as members of the MBW complex in regulating both anthocyanin and proanthocyanin accumulation (Gonzalez et al. 2008; Shi and Xie 2014; Xu et al. 2015). To investigate whether *SbC1* recruits the identified Tan1 and Tan2 to regulate anthocyanin accumulation in sorghum coleoptiles, we performed a yeast two-hybrid (Y2H) assay. As expected, physical interactions between *SbC1* and Tan2 were not observed in yeast (Fig. 5a). However, the *SbC1* protein physically interacted with Tan1 in the yeast system (Fig. 5a). We subsequently performed a bimolecular fluorescence complementation (BiFC) assay to confirm the interaction between *SbC1* and Tan1 in tobacco. Only tobacco leaves cotransformed with the *SbC1*-nYEP and Tan1-cYFP vectors presented a GFP signal localized

within the cell nucleus (Fig. 5b). Furthermore, we performed a Split-LUC assay in tobacco to confirm the interactions of *SbC1* and Tan1. The robust fluorescence signal was detected between *SbC1*-cLUC and Tan1-nLUC (*sbcl-a* as the negative control) (Fig. 5c-d). Additionally, a Co-IP assay was conducted to investigate the interaction between *SbC1* and Tan1 in tobacco plants. Flag-Tan1 and *SbC1*-GFP or GFP proteins were transiently expressed in tobacco leaves, followed by immunoprecipitation of Flag-Tan1 using anti-Flag antibody-conjugated agarose. *SbC1* and its interacting proteins were identified by immunoblot analysis with anti-Flag and anti-GFP antibodies. The result demonstrated that *SbC1*-GFP but not GFP itself can be coimmunoprecipitated by Tan1, implying that *SbC1* interacts with Tan1 in *planta* (Fig. 5e). To investigate the potential involvement of Tan1 in anthocyanin accumulation in sorghum coleoptiles, we determined the expression pattern of Tan1 in different sorghum tissues. The results revealed that Tan1 also presented a relatively high expression level in sorghum coleoptiles (Figure S5a). Considering that *SbC1* can physically interact with Tan1, Tan1 may also participate in the regulation of anthocyanin accumulation in sorghum coleoptiles through interactions with *SbC1*.

Transcriptome analysis revealed genes that were differentially expressed between HYZ and QKY

To elucidate the regulatory mechanism underlying colour pigmentation in sorghum coleoptiles, we conducted transcriptomic analyses on RNA samples extracted from the coleoptiles of two genetically similar landraces, HYZ (*SbC1*), with a red coleoptile, and QKY (*sbcl-a*), with a

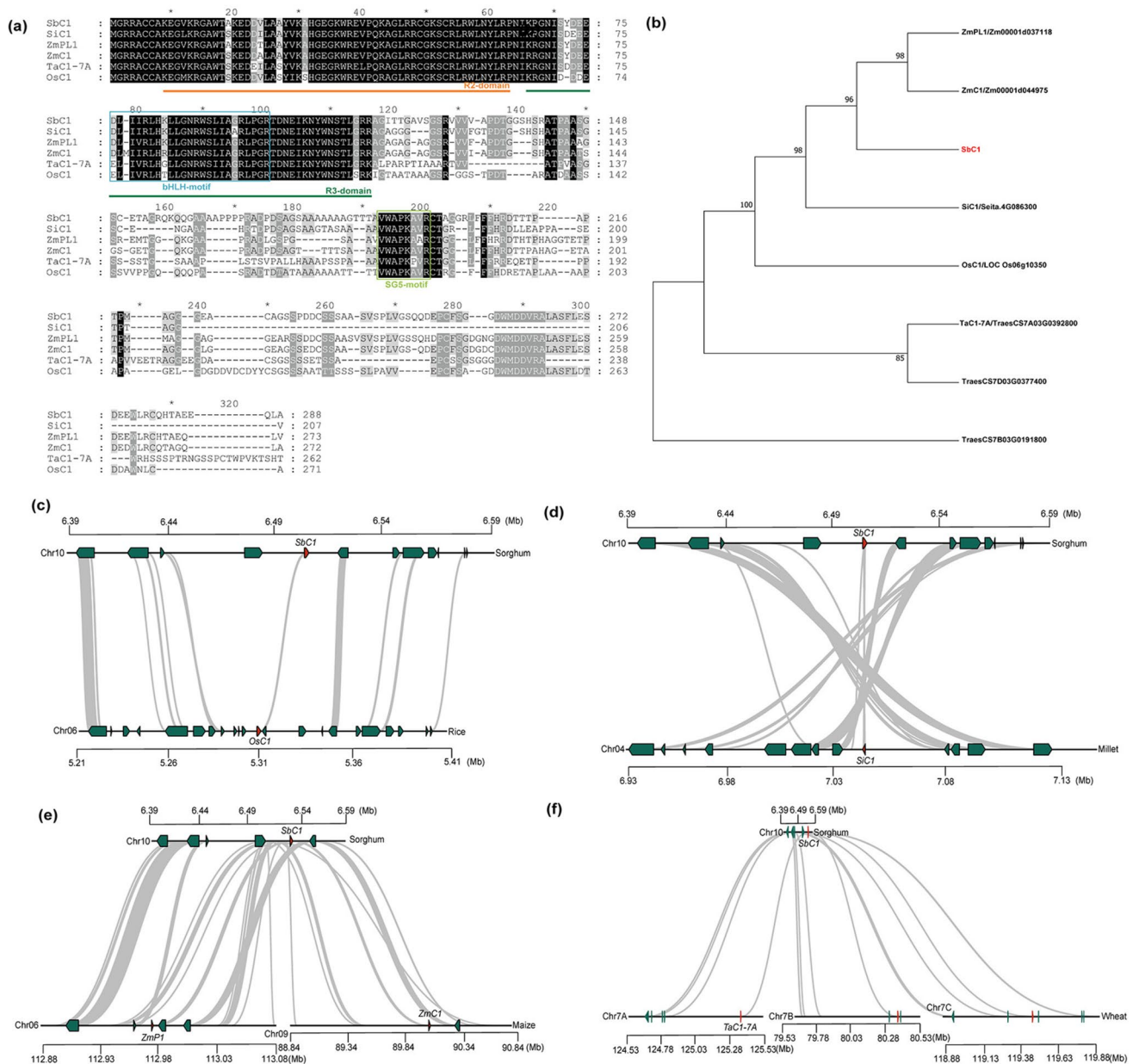


Fig. 4 Comparative genomic analysis of the C1 locus across different cereal species. **a** Protein sequence alignment of SbC1 with the homologous genes ZmC1, ZmPL1, OsC1, SiC1, and TaC1-7A, which revealed conserved motifs, including the bHLH binding motif and subgroup 5 characteristic motif. **b** Phylogenetic tree showing the evolutionary relationship between SbC1 and its homologues in other cereal species, indicating SbC1's close relationship with ZmC1 and

ZmPL1. **c-f** Synteny analysis of the C1 locus across different cereal species, illustrating the conserved genomic regions surrounding the C1 locus compared with rice (c), foxtail millet (d), maize (e), and wheat (f). The grey lines represent conserved syntenic blocks, highlighting the evolutionary conservation of the C1 locus among these species

green coleoptile (Sequencing results revealed that QKY and HYZ share 99.5% of their genetic loci across 2,015,850 polymorphic SNP sites, with no genetic variation detected in any anthocyanin biosynthesis-related genes, except for the *C1* gene). After data filtration, the clean reads were mapped to the reference genome of T2T-HYZ (Ding et al. 2024). In total, we identified 1573 differentially expressed genes

(DEGs), comprising 906 upregulated and 667 downregulated genes. These DEGs were mapped to 98 KEGG pathways. Enrichment analysis revealed that allelic variation in *SbC1* primarily influenced genes involved in phenylpropanoid biosynthesis, MPPK signal transduction, and flavonoid biosynthesis (Fig. 6a). The GO analysis revealed that the identified DEGs were associated primarily with functions

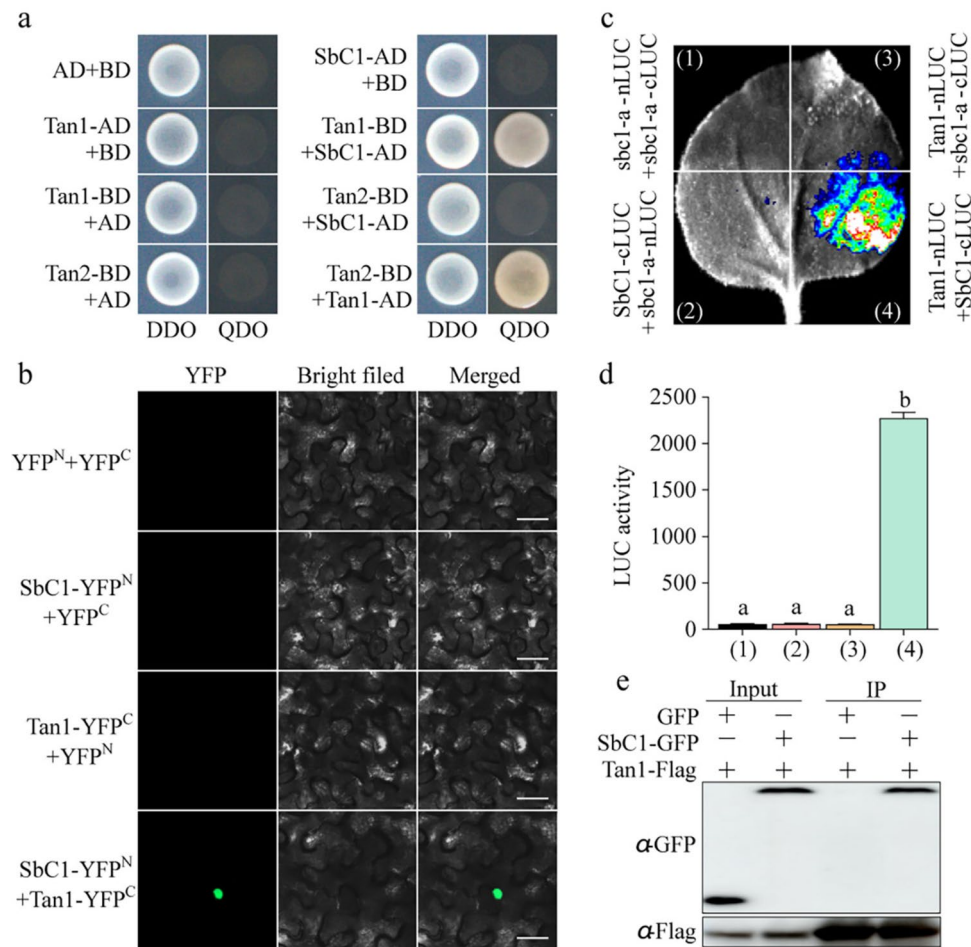


Fig. 5 SbC1 physically interacts with Tan1 in the nucleus. **a** Yeast two-hybrid (Y2H) assay for the interaction between SbC1 and Tan1. Yeast transformants were plated on minimal SC-Trp/Leu medium (DDO) or SC-Trp/Leu/His/Ade (QDO) medium, and photographs were taken after the transformants had grown for 3 days at 28 °C. The interaction between Tan1-AD and Tan2-BD as positive control. **b** Bimolecular fluorescence complementation (BiFC) assay. Various versions of SbC1 that interact with Tan1 were fused with the N- or C-terminus of yellow fluorescent protein (YFP) and transiently expressed in tobacco leaves for 3 days. The YFP fluorescence signal was detected via a Leica SP5 confocal microscope. bars=100 mm.

c Split-luciferase complementation (SLC) assay showing that SbC1 interacts with Tan1 in tobacco leaves. Protein *sbc1-a* was used as the negative control. **d** Quantification of LUC activity shown in (b). The error bars represent means ± SDs (n=3). Different lowercase letters indicate a significant difference at $P < 0.05$ on the basis of one-way analysis of variance (ANOVA). **e** Coimmunoprecipitation assay showing the physical interaction between SbC1 and Tan1 in vivo. CoIP was conducted with Flag-Trap beads, and protein complexes were probed by immunoblot analysis with anti-GFP and anti-Flag antibodies

related to oxidoreductase activity, iron ion binding, and peroxidase activity (Fig. 6b). Further analysis of differences in the expression of genes between the two varieties revealed a significant upregulation of genes involved in the early stage of anthocyanin biosynthesis (*CHS*, *CHI*, *F3H*, *F3'H*), as well as those involved in the late stage of anthocyanin synthesis (*DRF*, *ANS*, *UFGT*) specifically in the red coleoptiles of HYZ (Fig. 6c). To validate the reliability of the transcriptomic data, qRT-PCR was conducted on 8 genes involved in anthocyanin biosynthesis, revealing strong concordance with the RNA-seq results (Figure S7). Notably, the expression levels of *F3H*, *DFR*, *ANS*, and *UFGT* were negligible (FPKM < 0.1) (Fig. 6c) in QKY, suggesting that *SbC1* might

regulate anthocyanin biosynthesis in sorghum coleoptiles by controlling the expression of anthocyanin biosynthesis genes, especially *F3H*, *DFR*, *ANS* and *UFGT*.

SbC1 can complement the anthocyanin deficiency phenotype of the rice *Osc1* mutant

A previous study reported that the rice variety 9311 has green coleoptiles and leaf sheaths due to the null mutation of *Osc1* (Qiao et al. 2021). To further verify the function of *SbC1* in anthocyanin accumulation and investigate the conservation of gene functions among homologous genes, we introduced the coding sequences (CDSs) driven by a

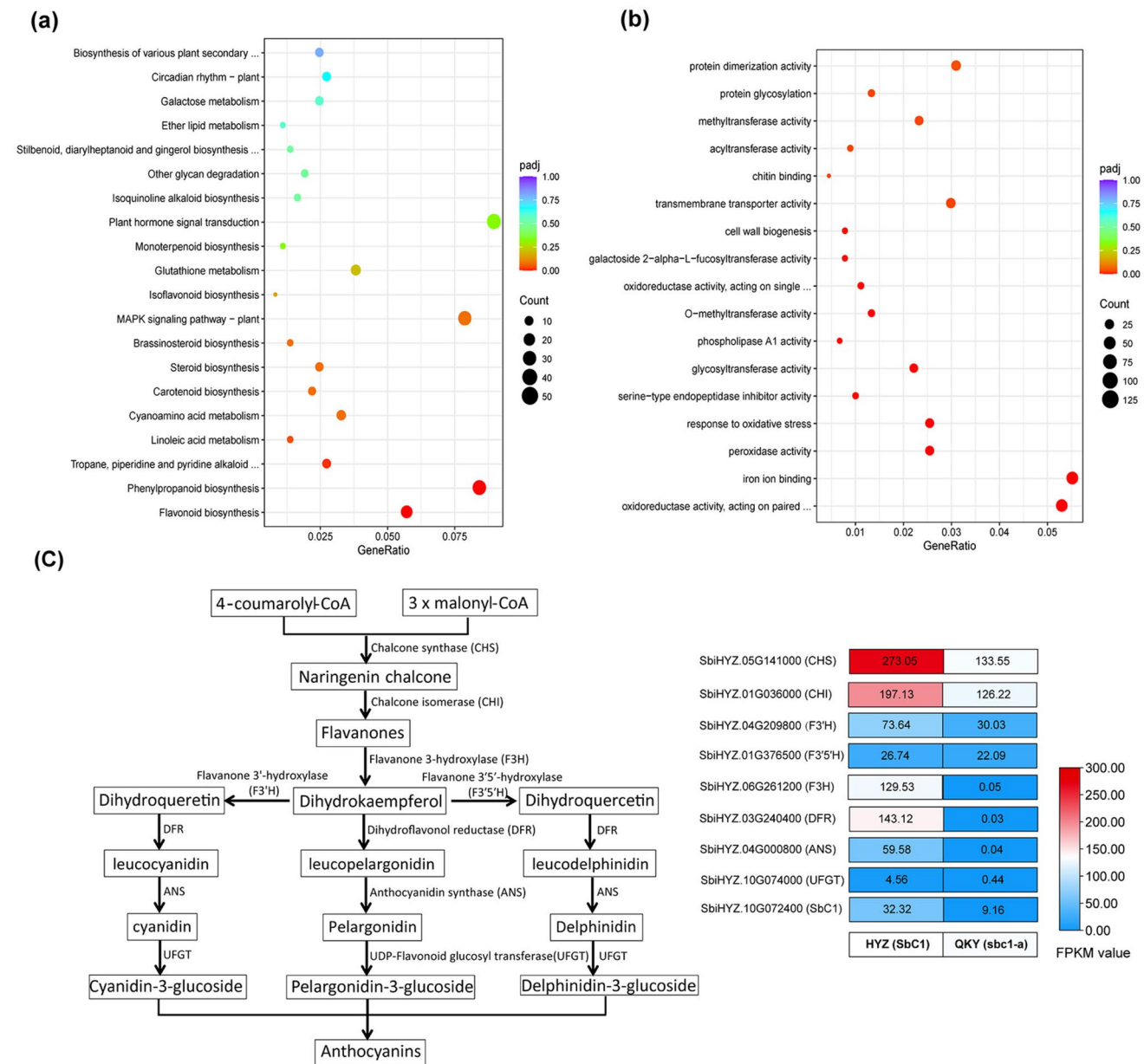


Fig. 6 RNA-seq analysis results. **a** Significantly enriched Kyoto Encyclopedia of Genes and Genomes pathways associated with the DEGs from HYZ (SbC1) vs. QKY (*sbc1-a*). **b** Significantly enriched

Gene Ontology terms associated with DEGs from HYZ (SbC1) vs. QKY (*sbc1-a*). **c** Anthocyanin biosynthesis pathway and expression patterns of related genes in HYZ (SbC1) and QKY (*sbc1-a*)

35S promoter into the rice variety 9311. Both T2 transgenic and 9311 plants were cultivated under controlled conditions in a greenhouse, and phenotypic identification revealed noticeable pigmentation in various vegetative tissues, including the leaf sheath, ligule, stigma, apiculus, and hull apex, of the transgenic rice compared with those of the wild-type plants (Fig. 7a-d), indicating that SbC1 plays an analogous function with OsC1 in terms of positive regulate anthocyanin accumulation.

SbC1 improves drought tolerance in rice and sorghum

Numerous studies have demonstrated that the accumulation of plant anthocyanins can be induced by various stresses, serving as a defensive mechanism and augmenting plant biotic stress tolerance, such as by increasing drought resistance (Zhang et al. 2014; Huang et al. 2019;

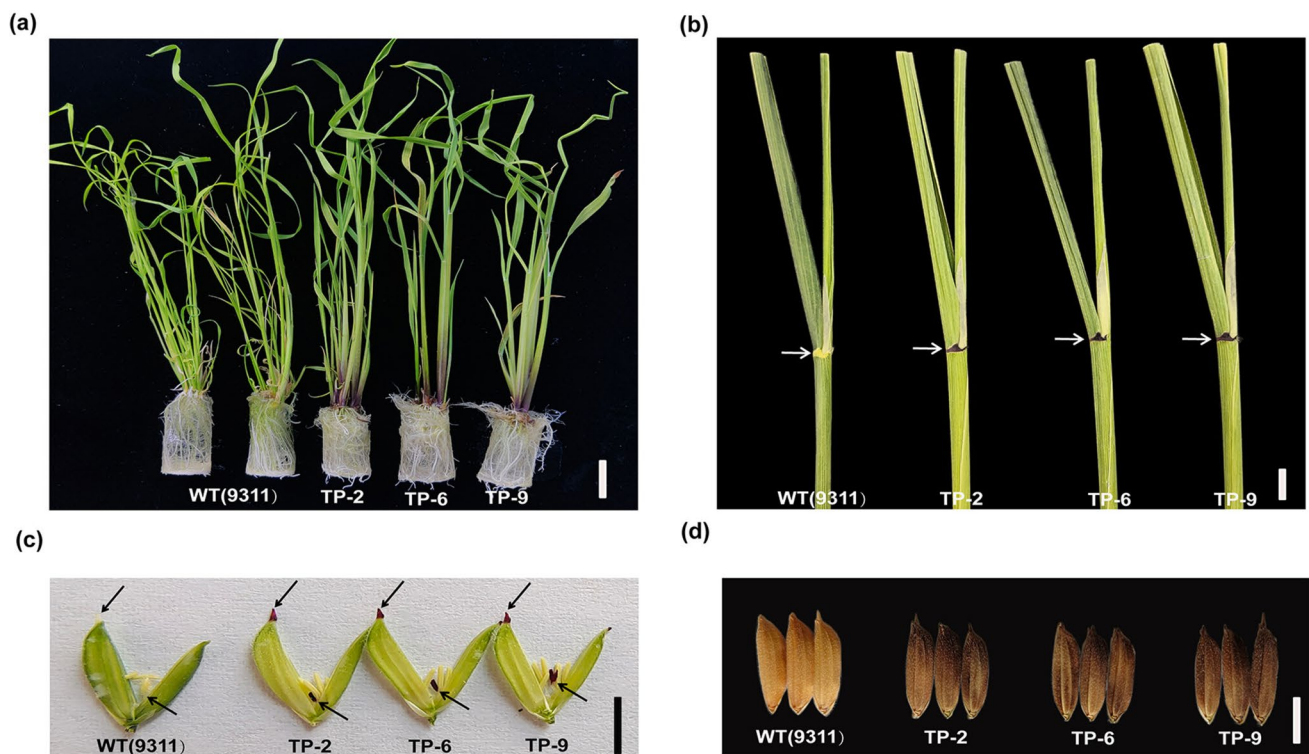


Fig. 7 Functional assessment of *SbC1* verified by overexpression in rice (9311) lines. **a** Overexpression of *SbC1* in rice 9311 lines; bars=1 cm. **b** Leaf phenotype comparison between the wild-type

and transgenic lines; bars=1 cm. **c–d** Spikelet and seed comparison between the wild-type and transgenic lines; bars=1 cm

An et al. 2020a, 2020b; Liao et al. 2022). Considering the substantial deposition of anthocyanins in various organs (Fig. 7a–d) resulting from the high expression of *SbC1* in transgenic rice, we hypothesized that these transgenic lines may exhibit robust resistance against multiple biotic stresses. To investigate the potential of *SbC1* overexpression to increase drought resistance in rice, we exposed 21-day-old transgenic and mutant rice plants to drought stress conditions. The results revealed no statistically significant differences in the growth or physical characteristics of the plants under standard irrigation with water (Fig. 8a). However, the fresh weight and survival rates of the transgenic lines overexpressing *SbC1* were greater than those of the mutants treated with PEG6000 (to simulate drought stress) (Fig. 8b, c). We subsequently conducted a similar drought stress treatment on the sorghum varieties HYZ and QKY. The fresh weight and survival rates of HYZ in the presence of functional *SbC1*, as depicted in Fig. 8d, were markedly greater than those of QKY, which is similar to the outcomes observed in transgenic rice (Fig. 8e, f). Taken together, these findings provide evidence for the potential of *SbC1* to enhance drought tolerance in both rice and sorghum, highlighting its pivotal role in increasing plant resilience under drought stress.

Discussion

SbC1 is a candidate gene in the Rs2 locus that controls coleoptile colour

Pigmentation of the coleoptile represents a readily distinguishable phenotypic trait that can be used to determine the homogeneity or specificity of varieties (Khlestkina et al. 2002). Early studies utilized molecular markers such as RFLP, AFLP, and SSR to develop genetic linkage maps for this trait, pinpointing the QTLs controlling sorghum coleoptile colour as Rs1 and Rs2. Within the R931945-2-2/IS 8525 genetic population, Rs1 was mapped between 108.1 and 109.5 centimorgans (cM) on chromosome 6 (Mace and Jordan 2010), corresponding to a physical location between 52.20 and 56.61 Mb in the BTx623 genome (v3.1.1) (Mace et al. 2019). Further research indicated that Rs1 might also influence overall sorghum plant colouration (Boyles et al. 2017). Rs2 was genetically mapped between 32.1 and 35.3 cM on chromosome 10, corresponding to a physical location between 3.28 and 5.95 Mb on the BTx623 genome (v3.1.1) (Mace and Jordan 2010).

In this study, we conducted a GWAS using 242 Chinese sorghum germplasms and successfully mapped the genomic region associated with coleoptile colour to chromosome 10,

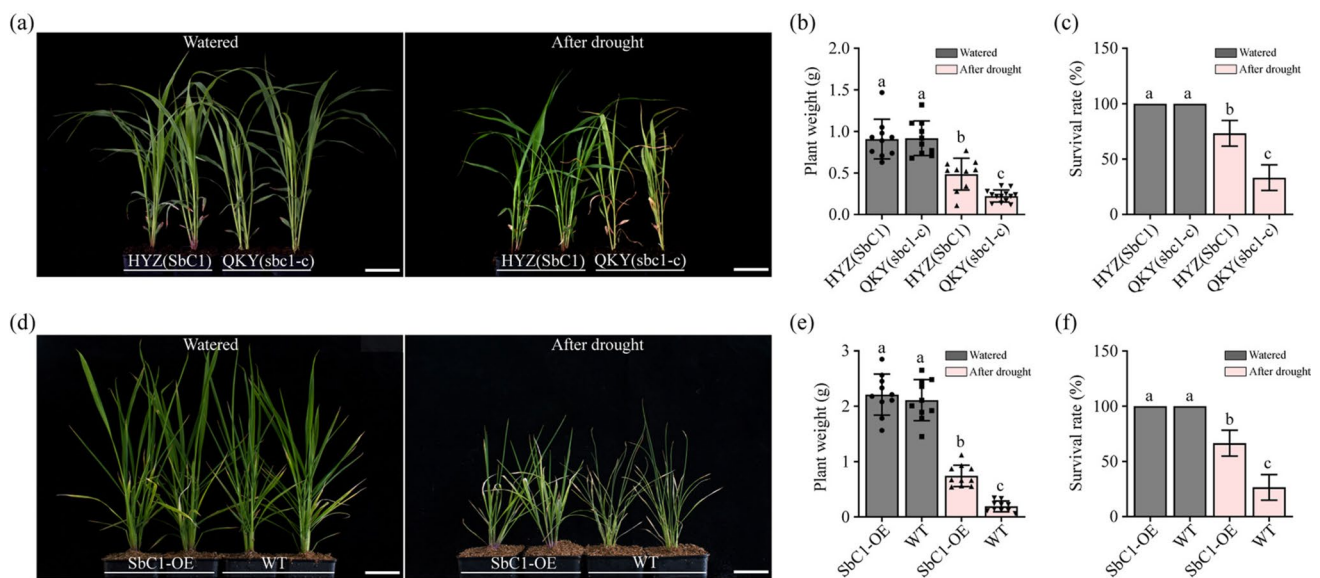


Fig. 8 Drought tolerance analyses. **a** Phenotypes of HYZ and QKY plants in response to drought stress. bars=2 cm. **b–c** Plant weight and survival rates of HYZ and QKY. **c** Phenotypes of 9311 (WT) and SbC1-overexpressing plants in response to drought stress. **e–f** Weight

and survival rates of 9311 (WT) and SbC1-overexpressing plants. bars=2 cm. The error bars represent means ± SDs (n=3). Different lowercase letters indicate a significant difference at $P < 0.05$ based on a t-test

which was consistent with the previously identified Rs2 locus. VIGS experiments further substantiated the pivotal role of *SbC1* in controlling sorghum coleoptile pigmentation. Comparative genomic analysis with other sorghum genomes (HYZ) identified *SbC1*, a gene absent from the BTx623 genome (v3.1.1). Moreover, genomic sequence analysis revealed that *SbC1* was incorrectly annotated in version v3.1.1, which hindered the accurate identification of this key gene through homology analysis (Li et al. 2022). The *SbC1* gene in the updated BTx623 reference genome (v5.1) (https://phytozome-next.jgi.doe.gov/info/Sbicolor_v5_1) is annotated as Sobic.010G077237. Comparative analysis between the *SbC1* genes from HYZ and BTx623 revealed sequence variations in both intronic and exonic regions, leading to alterations in the amino acid sequences (Figure S8). Whether these variations impact the function of the *SbC1* gene requires further experimental validation.

The loss of anthocyanins in cereals might share the same genetic basis

During the domestication and improvement of various cereals, many traits have undergone parallel changes (Huang et al. 2022), such as grain size (Tao et al. 2020, 2021a, 2021b). One significant observation across these major cereals is the loss of anthocyanin accumulation in plant organs, including anthers, seeds, leaves, stems, and roots. However, whether these parallel trait changes occur via a common

genetic mechanism remains largely unexplored (Li et al. 2022).

R2R3-MYB transcription factors are pivotal in anthocyanin synthesis, with numerous R2R3-MYBs identified as positive regulators of anthocyanin biosynthesis in cereals. *ZmC1*, the first cloned MYB transcription factor-encoding gene in maize, regulates anthocyanin biosynthesis in the grain aleurone layer (Cone et al. 1993). In this study, we identified the *SbC1* (R2R3-MYB) gene as a key activator that promotes anthocyanin biosynthesis in sorghum coleoptiles, with nucleotide mutations in its exons impacting gene function. *ZmC1*, *ZmP11*, *OsC1*, *SiC1*, and *TaC1-7A*, are orthologs of *SbC1*. Previous studies have shown that these genes have undergone various mutations during the domestication, diversification, and crop improvement processes, leading to the loss of gene function and subsequently affecting the accumulation of anthocyanins in the vegetative and floral organs of these crops (Morohashi et al. 2012; Jiang et al. 2018; Du et al. 2022; Li et al. 2022). Recent studies also identified another ortholog of *SbC1*, the R2R3-MYB transcription factor *TdRCA1*, which specifically regulates anthocyanin biosynthesis in the coleoptile of wild emmer wheat (Li et al. 2024). In this study, heterologous expression of *SbC1* in the rice *OsC1* mutant variety 9311 led to anthocyanin accumulation in various tissues, such as the leaf sheath, ligule, hull apex, and stigma, indicating that *SbC1* and *OsC1* share a conserved regulatory mechanism in anthocyanin biosynthesis. All these findings suggest that the *c1* genes responsible for anthocyanin production played

a key role in the loss of pigmentation across different cereal species.

The purple apiculus trait controlled by *OsCI* is common in wild rice but rare in cultivars (Saitoh et al. 2004; Choudhury et al. 2014). However, certain traits for crop improvement, such as grain texture and flavour, governed by genes such as *waxy* and *BADH2*, exhibit allelic diversity across various landraces and cultivars, reflecting regional preferences (Olsen and Purugganan 2002; Kovach et al. 2009). Similar phenomena were observed in our study, and the allelic diversity of *SbCI* revealed signatures of selection in Chinese sorghum but not in improved varieties in Asia and North America.

SbCI specifically promotes coleoptile anthocyanin accumulation by activating the expression of anthocyanin biosynthesis genes

Classic genetic analyses has shown that anthocyanin and tannin synthesis in sorghum grains is controlled by a pair of recessive epistatic genes (*B₁/Tan1* and *B₂/Tan2*) (Smith et al. 2000). In this study, through VIGS, haplotype analysis, and phenotypic investigation, we confirmed that allelic variation in the *SbCI* gene affects anthocyanin accumulation in coleoptiles but not in grains. MYB transcription factors typically determine the specific activation of target genes (Li 2014; Liu et al. 2015; Allan and Espley 2018). In this study, we found that allelic variation in *SbCI* affects the expression of structural genes related to anthocyanin biosynthesis, such as F3H, DFR, ANS, and UFGT. These findings are consistent with those of previous studies, suggesting that *SbCI* may directly bind to the promoters of structural genes related to anthocyanin biosynthesis, activating their transcription and thereby regulating anthocyanin accumulation in coleoptiles. Therefore, future research should focus on exploring how *SbCI* can be utilized in breeding programs to enhance anthocyanin accumulation and improve stress resistance in sorghum and other cereal crops.

The components of the MBW complex responsible for anthocyanin accumulation in sorghum coleoptiles remain elusive

The MBW ternary complex, comprising MYB, bHLH, and WD40 transcription factors, is critical in the biosynthesis of anthocyanins and proanthocyanidins across various plant species (Dixon et al. 2013). The bHLH transcription factor plays a crucial role in facilitating the assembly of transcriptional complexes at the promoters of anthocyanin biosynthetic genes through its interaction with the R3 region of its R2R3-MYB partner (Xu et al. 2015; LaFountain and Yuan 2021). It is widely accepted that WD40 proteins function primarily as scaffold proteins that facilitate interactions

between bHLH proteins and various MYB proteins to regulate anthocyanin accumulation in different tissues rather than directly interacting with MYB proteins (Gonzalez et al. 2008; An et al. 2012; Xu et al. 2015; Gao et al. 2018; Liu et al. 2021). However, recent studies have reported direct interactions between WD40 and MYB proteins. For example, RcTTG1 interacts with RcMYB1, OsTTG1 with OsC1, FaLWD1 with FaMYB5, and RhWD40 with RhMYB114a and RhMYB3b (Yang et al. 2021; Jiang et al. 2023; Yan et al. 2023; He et al. 2023a). In this study, we conducted an in-depth investigation of the interactions between WD40, bHLH, and MYB transcription factors, with a particular focus on their regulatory roles in anthocyanin accumulation within sorghum coleoptiles and grains.

In our study, via Y2H, BiFC, and Co-IP assays, we determined that the sorghum WD40 protein Tan1 interacts with *SbCI*. However, *SbCI* did not interact with Tan2, a bHLH transcription factor previously reported to be involved in sorghum proanthocyanin accumulation. Our group and Wu et al. reported that the Tan2 gene regulates anthocyanin and tannin biosynthesis in sorghum grain (Wu et al. 2019; Zhang et al. 2023). Analysis of gene expression databases across different tissues and stages of tannin sorghum revealed that *Tan2* is not expressed in coleoptiles at the five-leaf stage, when coleoptile colouration is most prominent, but is highly expressed during grain development (Figure S5b). Thus, we hypothesize that Tan2 is not involved in regulating anthocyanin synthesis in coleoptiles but regulate the synthesis of anthocyanins and proanthocyanidins in grains through interactions with Tan1.

In maize, *ZmPac1* is essential for anthocyanin accumulation in the aleurone layer and scutellum, whereas anthocyanin synthesis in the pericarp and other tissues is unaffected in *ZmPac1* mutants (Carey et al. 2004; Chatham et al. 2019). In rice, a mutation in *OsTTG1* (Tan1 homologue) significantly reduced anthocyanin accumulation in all tissues (Yang et al. 2021), indicating that regulatory genes for anthocyanin biosynthesis are relatively conserved among different Gramineae species, although their regulatory mechanisms exhibit specificity. Intriguingly, studies by Wu et al. (2019) and phenotypic data from 479 global sorghum germplasm resources revealed that functional variation in *Tan1* affects grain anthocyanin and tannin biosynthesis but not coleoptile anthocyanin biosynthesis (Figure S9). These findings suggest the existence of other functionally redundant WD40 proteins involved in anthocyanin biosynthesis in coleoptiles.

The application of *SbCI* in stress resistance

Anthocyanins provide protection to crops under various biotic and abiotic stresses (Das et al. 2012; An et al. 2020a, 2020b). For example, anthocyanins in wheat

seeds and coleoptiles promote root and shoot elongation under osmotic stress conditions (Shoeva et al. 2017). In this study, compared with wild-type rice, transgenic rice demonstrated higher fresh weight and survival rates after drought stress. Similarly, the sorghum HYZ (carrying the *SbC1*) showed advantages in drought tolerance over the QKY (carrying the *sbc1-a*). This enhanced drought tolerance is likely due to the formation of an anthocyanin-rich flavonoid barrier in the coleoptiles of seedlings, which mitigates damage caused by drought stress. Given that transgenic rice lines continuously express the *SbC1* gene during the mature stage and exhibit pigmented plants, it is possible that they may also have stress resistance at later developmental stages; however, this still requires further validation through field trials. Similarly, overexpression of *TdRCA1*, a homolog of *SbC1*, not only increases anthocyanin accumulation in wheat coleoptiles but also significantly enhances seedling resistance to Fusarium crown rot (Li et al. 2024). These results suggest that the accumulation of anthocyanins in coleoptiles may enhance the ability of seedlings to cope with stress conditions, and also indicate the potential application of the *SbC1* gene in improving crop stress resistance.

In summary, Through GWAS, VIGS experiments, and haplotype analysis, we successfully identified an R2R3-MYB transcription factor, *SbC1*, specifically positive regulates anthocyanin accumulation in sorghum coleoptile. Transcriptome analysis further demonstrated the activation of anthocyanin biosynthesis genes by *SbC1*. Additionally, comparative genomic analysis and transformation experiments have proved the analogous function of *SbC1* to *OsC1* in terms of regulating anthocyanin biosynthesis. Additionally, the interaction between *SbC1* and *Tan1* was also demonstrated using different strategies. Recently, Yang et al. identified a bHLH transcription factor, *SbPLSH1*, that is closely associated with anthocyanin accumulation in the sorghum leaf sheath (Yang et al. 2024). Whether *SbPLSH1* interacts with *SbC1* or whether other unidentified bHLH members participate in coleoptile anthocyanin accumulation remains to be further elucidated. In future studies, we will focus on the identification and function analysis of other genes that regulate sorghum anthocyanin and tannin accumulation to comprehensively revealing the regulatory roles of these genes in coleoptile and other tissues pigmentation in sorghum.

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Data availability The raw sequence data presented in this paper were previously published (Boatwright et al. 2022; Zhang et al. 2023) and have been deposited in both the China National GeneBank (CNP0002968) and the European Variant Archive (PRJEB51985).

Declarations

Conflict of interest The authors declare no conflict of interest.

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